

Simpson 260 Series 8

DIY Calibration Procedure — The Config Bench | Rev 4b

John King | @TheConfigBench | Reference: Owon XDM1241

AC Station: Isolation Transformer → Dim Bulb Tester → Variac (0-130VAC) → APC UPS 1500 Step-Up

Revision History:

- Rev 1 — Base calibration procedure, wall power + XDM1241.
- Rev 2 — Isolation Transformer + Variac (0-130VAC). Documented 130VAC limit.
- Rev 3 — APC UPS 1500 reverse step-up. Full range to ~975VAC.
- Rev 4 — Dim Bulb Tester build + Transformer pre-test procedure added.
- Rev 4b — Table font size reduced and cells wrapped — improved readability.

Wall 120VAC → [Isolation Transformer] → [Dim Bulb Tester] → [Variac 0–130VAC] → [UPS Step-Up if needed] → [XDM1241 monitoring] → [Simpson 260 Under Test]

Section 1 — Dim Bulb Tester: Theory, Build & Parts List

The Dim Bulb Tester places an incandescent bulb IN SERIES with the load. Under normal operation the bulb glows dimly or not at all — the VOM under calibration draws microamps. On a fault or short, current spikes, the bulb flares bright and absorbs the fault current instead of destroying your equipment, Variac winding, or tripping a breaker. Placed between the Isolation Transformer and Variac it protects everything downstream.

Reading the Bulb:

Bulb Behavior	What It Means	Action
Off / barely glows	Normal — VOM calibration draws near-zero current	Proceed normally
Glows dimly	Small load present — acceptable	Monitor — normal for resistive loads
Glows ~50% brightness	Moderate current — investigate if unexpected	Check circuit before proceeding
Full brightness	HIGH current — likely fault or short	Power down immediately — investigate
Flashes bright then dims	Capacitor charging — common in power supplies	Normal — continue monitoring

Build — Parts List:

Qty	Part	Specification / Notes	Source
1	Enclosure	Metal or heavy plastic project box — 4"x4"x2" min	Home Depot / Amazon
1	Porcelain light socket	Medium base — 660W rated minimum. NOT plastic — bulbs run extremely hot	Home Depot / electrical supply
1	Incandescent bulb	100W for sensitive fault detection. 150–250W for higher current work. Keep both on bench.	Hardware store
1	Clear acrylic viewing window	Must see bulb brightness from outside — cut to fit enclosure	Hardware store
1	SPST bypass toggle switch	15A–20A rated, 120VAC — wired in PARALLEL across bulb socket	Electronics supply / Amazon
1	SPST power switch (optional)	Main on/off — 15A rated	Electronics supply

Qty	Part	Specification / Notes	Source
1	IEC inlet or AC inlet	Input from isolation transformer output	Amazon / Mouser
1	Standard duplex AC outlet	Output to Variac	Home Depot
1	Panel LED indicator (optional)	120VAC rated panel lamp — shows unit powered	Electronics supply
—	12 AWG wire	All internal wiring — 14 AWG minimum acceptable	Home Depot
—	Wire nuts or terminal strip	Secure all connections — no bare splices	Home Depot

Wiring — Series Circuit:

HOT path (series through bulb): AC Inlet HOT → Power Switch (opt.) → Bypass Switch terminal 1 → Bulb Socket HOT → Bypass Switch terminal 2 → AC Outlet HOT
 NEUTRAL path (straight through — does NOT pass through bulb or switch): AC Inlet NEUTRAL → AC Outlet NEUTRAL
 GROUND path (straight through): AC Inlet GROUND → Enclosure → AC Outlet GROUND
 Bypass Switch wired IN PARALLEL across bulb: OPEN = bulb in circuit = protection ACTIVE
 CLOSED = bulb bypassed = full power downstream (only after confirming no fault)
 Label clearly: "BYPASS OPEN = PROTECTION ACTIVE"

■ ■ Use porcelain socket ONLY — plastic melts. 12 AWG wiring throughout. Viewing window must allow bulb observation without opening enclosure.

Section 2 — Transformer Pre-Test Procedure

Perform this sequence on ANY transformer before placing it in the calibration chain — especially salvage units. Tests proceed from unpowered (safe) to powered (through Dim Bulb Tester).

■ ■ Unpowered tests first. Then powered tests through Dim Bulb Tester at reduced voltage only. Never apply full line voltage to an untested salvage transformer directly.

Test 1 — Visual Inspection (Unpowered)

- Check all winding leads for cracked or brittle insulation.
- Check core for overheating signs: darkened varnish, burnt smell, discoloration.
- Verify core laminations are tight — loose laminations cause excessive hum.
- Check for previous overload: bulging, melted insulation, scorch marks.
- **Any of the above found — do not proceed. Discard or rewind.**

Test 2 — Winding Continuity (Unpowered — XDM1241 Resistance Mode)

Test	Meter Leads	Expected Result	Fail Condition
Primary winding continuity	Primary lead 1 → Primary lead 2	Low Ω — heavy wire. Typical: 2–15 Ω on large transformer. Record value: _____ Ω	OL = open winding Near 0 Ω = shorted turns
Secondary winding continuity	Secondary lead 1 → Secondary lead 2	Moderate Ω — higher than primary. Typical: 5–50 Ω . Record value: _____ Ω	OL = open winding Near 0 Ω = shorted turns
Primary-to-secondary isolation	Any primary lead → any secondary lead	OL (infinite) — windings must be fully isolated from each other	Any finite reading = insulation failure — do not use

Test 3 — Winding-to-Core Insulation (Unpowered — XDM1241 Resistance Mode)

Test	Meter Leads	Expected Result	Fail Condition
Primary to core/frame	Any primary lead → transformer core or metal frame	OL (infinite resistance)	Any reading = insulation breakdown — do not use
Secondary to core/frame	Any secondary lead → transformer core or metal frame	OL (infinite resistance)	Any reading = insulation breakdown — do not use

■ A megohmmeter (megger) is the professional standard for insulation testing — it tests at 500V or 1000V DC to stress the insulation properly. A DMM resistance test at battery voltage is a valid initial screen but cannot substitute for a megger on HV applications. Future @TheConfigBench episode topic.

Test 4 — No-Load Energization (Powered — Through Dim Bulb Tester)

■ ■ First powered test. Dim Bulb bypass switch OPEN. Secondary leads left open — nothing connected to output.

1 Setup

Wall → Isolation Xfmr → Dim Bulb Tester (bypass OPEN) → transformer PRIMARY leads. Secondary open.

2 Power On — Observe Bulb

Bulb should glow very dimly or not at all — transformer draws only magnetizing current at no load (~50–200mA). Bright bulb = internal short. Power down immediately.

3 Listen

Quiet 60Hz hum is normal. Loud buzzing, rattling, or clicking = loose laminations or winding problem.

4 Smell

After 2 minutes — any burning smell or acrid plastic odor = insulation breaking down. Power down immediately.

5 Measure Primary Voltage

XDM1241 across primary input — confirms voltage applied. Record: _____ VAC

6 Measure Secondary Voltage

Move XDM1241 to secondary output leads. Record no-load secondary voltage: _____ VAC. This is your turns ratio measurement.

Test 5 — Light Load Test (Powered — Through Dim Bulb Tester)

1 Calculate Load

For ~16VAC secondary: 68Ω 10W resistor gives ~235mA load. Or use a 12V automotive bulb across the secondary.

2 Connect Load

Power OFF first. Connect load resistor or bulb across secondary output.

3 Power On — Observe Dim Bulb

Slight increase in Dim Bulb brightness is normal — transformer now delivering current. Full brightness = problem.

4 Measure Secondary Under Load

XDM1241 across secondary with load. Compare to no-load reading. Drop of 1–3V is normal regulation. Drop >10% = marginal transformer.

5 5-Minute Thermal Check

Run 5 minutes. Transformer should be warm — not hot. Uncomfortably hot = excessive loss, possible shorted turns. Power down and investigate.

Test 6 — Turns Ratio Measurement

- During Test 4 no-load: measure PRIMARY voltage with XDM1241 → record exact value.
- Move XDM1241 to SECONDARY leads → record exact secondary voltage.
- **Turns Ratio = Primary Voltage ÷ Secondary Voltage**
- Example: 120.4VAC ÷ 16.1VAC = **7.48:1**
- Use this ratio for all Variac input target calculations in Section 4.
- Record permanently in calibration record sheet.

Transformer Pre-Test Checklist:

Test	Method	Pass Criteria	Your Result
1. Visual inspection	Eyes	No burns, cracked insulation, loose core	■ Pass ■ Fail
2a. Primary continuity	XDM1241 Ω	Low Ω — not OL, not near 0	_____ Ω
2b. Secondary continuity	XDM1241 Ω	Moderate Ω — not OL	_____ Ω
2c. Primary-to-secondary isolation	XDM1241 Ω	OL (infinite)	■ Pass ■ Fail
3a. Primary-to-core insulation	XDM1241 Ω	OL (infinite)	■ Pass ■ Fail
3b. Secondary-to-core insulation	XDM1241 Ω	OL (infinite)	■ Pass ■ Fail
4. No-load energization	Dim Bulb + XDM1241	Bulb dim/off, no smell, quiet hum	■ Pass ■ Fail
5. Light load test	Dim Bulb + resistive load	Slight Dim Bulb increase, <3V regulation drop, warm not hot	■ Pass ■ Fail
6. Turns ratio measured	XDM1241 AC V	Calculated and recorded	_____ : 1

■ All tests passed — transformer characterized, confirmed safe, ready for calibration chain. Label leads and record turns ratio.

Section 3 — Extended Range Transformer Selection

Attribute	Small 10:1 Consumer Transformer	APC UPS 1500 Transformer (Preferred)
VA Rating	5–20VA typical	Designed for 1500VA continuous operation
Wire Gauge	Fine gauge windings	Heavy gauge — built for real sustained current
Insulation Rating	Rated for 12VAC secondary — marginal at HV reverse operation	Rated for 120VAC primary — robust insulation margins throughout
Thermal Capacity	Heats up under sustained use	Massive thermal mass — barely notices calibration loads
Max Safe HV Output	~500V — insulation risk above this	1000V+ safely achievable with proper pre-test
Pre-Test Required?	Yes — full Section 2 procedure	Yes — full Section 2 procedure
Recommended Use	Fallback only if UPS transformer unavailable	■ PRIMARY extended range transformer

Section 4 — Turns Ratio Variac Input Calculator

Formula: Variac Input = Target Output Voltage ÷ Turns Ratio | Measure YOUR ratio in Section 2 Test 6. Fill in "Your Setting" column.

Simpson AC Range	Target Test Voltage	Variac Input (7.5:1 example)	Your Variac Setting (fill in)	Source / Method
2.5V	1.25V (mid-scale)	0.17 VAC	_____	Variac direct — no step-up
10V	8V (80% scale)	1.1 VAC	_____	Variac direct — no step-up
50V	40V (80% scale)	5.3 VAC	_____	Variac direct — no step-up
250V	125V (mid-scale)	16.7 VAC	_____	Variac → UPS step-up
500V	250V (50% scale)	33.3 VAC	_____	Variac → UPS step-up
1000V	500V (50% scale)	66.7 VAC	_____	Variac → UPS step-up
1000V max	~975V (130VAC in)	130 VAC max	130 VAC	Variac → UPS step-up — 97.5% range
5000V	Not achievable	333 VAC needed	N/A	■ Beyond Variac — do not attempt

■■ 1000V full-scale requires ~133VAC input — 3VAC over your 130VAC Variac limit. Dial to 130VAC max → ~975VAC output. Perfectly acceptable — note as "verified to 975VAC" in calibration record.

Section 5 — Equipment List

Item	Your Equipment	Purpose
Reference Meter	Owon XDM1241	Primary reference standard — all measurements
Isolation Transformer	Bench isolation transformer	Safety — breaks earth ground. Always first in chain.
Dim Bulb Tester	Built per Section 1 — 100W bulb recommended	Fault current protection for all downstream equipment
Variac	0–130VAC variable autotransformer	Variable AC voltage source
Extended Range Transformer	APC UPS 1500 — reverse fed (pre-tested per Section 2)	Step-up to 130–975VAC output range
Fallback Transformer	Small 10:1 — reverse fed (pre-tested)	Backup only — max ~500V safely
DC Voltage Source	Regulated bench supply or fresh 9V battery	DC voltage and current calibration

Item	Your Equipment	Purpose
Precision Resistors	1% tolerance — 100Ω, 1kΩ, 10kΩ, 100kΩ	Ohms range verification
Current Limit Resistors	Calculated per Section 9 — see table	Safe DC current range testing
Fresh Batteries	1.5V AA + 9V	Ohms function power — install before calibrating
Small Screwdriver	Flat blade jeweler size	Mechanical zero + internal pot adjustment

Section 6 — Pre-Calibration Inspection

1 Visual Inspection

Check meter face, needle, glass, test lead jacks. Inspect battery compartment — clean with IPA and fiberglass scratch pen if needed.

2 Mechanical Zero

No leads connected. Meter flat on bench. Needle on '0' at left of scale. Adjust with small flat screwdriver in front-face zero slot. Very sensitive — move slowly.

3 Fresh Batteries

Replace 1.5V AA and 9V before calibrating. Old batteries cause ohms drift. Clean contacts if corroded.

4 Lay Flat on Back

Meter flat on its back for ALL calibration. D'Arsonval movement is gravity-sensitive. Per US Army TB and EEVblog procedure.

5 Warm Up XDM1241

Power on — 5 minutes minimum before use as reference standard.

6 Bench Chain Verify

Isolation Xfmr → Dim Bulb Tester (bypass OPEN) → Variac at zero → all connections solid before applying power.

Section 7 — Internal Adjustment Potentiometers

Remove back cover. Four pots accessible. Calibrate in order — R1 always first.

Pot	Function	Affects	Calibrate Order
R1	Main movement shunt — sets full scale sensitivity for entire meter	ALL DC and AC ranges	1st — always first
R2	Current range calibration	mA current ranges only	2nd
R25	AC low range calibration	2.5V AC range	3rd
R22	AC high range calibration	250V AC range	4th

■ EEVblog (kj7e, 2018): Adjust R1 for 40μA — 80% deflection on the 50μA range. Best linearity across all ranges. Then R2 for 800μA on the 1mA range.

Section 8 — DC Voltage Calibration

Use regulated DC bench supply. Variac/transformers are AC only. Set R1 first.

Range	Apply (DC Supply)	XDM1241 Reads	Simpson Target	Tolerance	Pass / Fail
2.5V	2.0V DC	Actual V	~2.0V	±3%	
10V	9V fresh battery	Actual V	~9.0V	±3%	
50V	40V DC bench supply	Actual V	~40V	±3%	
250V and above	Skip — HV DC not practical for home bench	—	—	N/A	N/A

Section 9 — DC Current Calibration

■■ NEVER connect current ranges directly across a voltage source. Always use a series resistor. XDM1241 in series to verify actual current BEFORE reading Simpson.

Range	Series Resistor (9V source)	Calculated Current	Internal Adjustment	XDM1241 in Series?
50 μ A	180 k Ω 1%	~50 μ A	R1 if needed	Yes — verify actual first
1 mA	9.1 k Ω 1%	~989 μ A	R2	Yes — verify actual first
10 mA	910 Ω 1%	~9.9 mA	None	Yes — verify actual first
100 mA	91 Ω 1%	~98.9 mA	None	Yes — verify actual first
500 mA	Bench supply — current limit to 400 mA	~400 mA	None	Yes — verify actual first

Section 10 — Resistance Calibration

Zero the meter on EACH range before measuring. Mid-scale is always most accurate. No internal ohms pot — fresh batteries are everything.

Range	Test Resistor	XDM1241 Actual Reading	Simpson Target	Most Accurate Region
R $\times$ 1	100 Ω 1%	Actual Ω	~100 Ω	Mid-scale ~20 Ω
R $\times$ 10	1 k Ω 1%	Actual Ω	~1 k Ω	Mid-scale ~200 Ω
R $\times$ 100	10 k Ω 1%	Actual Ω	~10 k Ω	Mid-scale ~2 k Ω
R $\times$ 1K	100 k Ω 1%	Actual Ω	~100 k Ω	Mid-scale ~20 k Ω
R $\times$ 100K	1 M Ω 1%	Actual Ω	~1 M Ω	Mid-scale ~2 M Ω

Section 11 — AC Voltage Calibration

Low ranges (2.5V–130V): Variac direct. Extended ranges (130V–975V): UPS step-up transformer. XDM1241 monitors output at ALL times.

R25 — AC Low Range (2.5V range):

- Variac direct — no step-up needed. Dial to **1.25VAC** (mid-scale target).
- XDM1241 monitors actual output voltage in parallel with Simpson.
- Set Simpson to **2.5V AC range**. Adjust **R25** until Simpson matches XDM1241.

- Verify at 2.0VAC — should track XDM1241 within $\pm 3\%$.

R22 — AC High Range (250V range) — UPS Step-Up:

- Switch to UPS transformer step-up chain. Variac output → UPS transformer secondary → HV output from primary.
- XDM1241 monitors HV output. Set Simpson to **250V AC range**.
- Dial Variac to calculated input for 125VAC output (mid-scale) — e.g. 16.7VAC for 7.5:1 ratio.
- Adjust **R22** until Simpson reading matches XDM1241.
- Verify at maximum achievable voltage — Simpson should track within $\pm 3\%$.

Full AC Range Verification Table:

Range	Source	Variac Input (7.5:1)	HV Output Target	Pot	Testable?
2.5V	Variac direct	1.25 VAC	1.25 VAC	R25	■ Full range
10V	Variac direct	8 VAC	8 VAC	—	■ Full range
50V	Variac direct	40 VAC	40 VAC	—	■ Full range
250V	UPS step-up	16.7 VAC	~125 VAC (mid-scale)	R22	■ Full — mid-scale ideal
500V	UPS step-up	33.3 VAC	~250 VAC (50% scale)	—	■ 50% scale — acceptable
1000V	UPS step-up	~130 VAC max	~975 VAC (97.5%)	—	■ 97.5% — note in record
5000V	N/A	Would need 333 VAC	—	—	■ Not verifiable — beyond Variac

Section 12 — Calibration Record Sheet

Print one copy per meter. Complete during calibration session. Store with the meter permanently.

Meter Serial / ID	_____	Date Calibrated	_____
Calibrated By	John King — @TheConfigBench	Reference Meter	Owon XDM1241
AC Low Source	Isolation Xfmr + Dim Bulb + Variac (0-130VAC)	AC High Source	Variac + APC UPS 1500 Xfmr (reverse fed)
DC Source	Bench Supply / 9V Battery	UPS Xfmr Ratio Measured	_____ : 1
Xfmr Pre-Test Passed?	■ Yes ■ No	Battery Replaced?	■ Yes ■ No
Mech Zero Adjusted?	■ Yes ■ No	Dim Bulb Tester Used?	■ Yes ■ No

Function	Range	Applied Value (XDM1241)	Simpson Reading	Error %	Pass / Fail
DC Voltage	2.5V				
DC Voltage	10V				
DC Voltage	50V				
DC Current	50 μ A				
DC Current	1 mA				
DC Current	10 mA				
Resistance	R $\times$ 1 (100 Ω)				
Resistance	R $\times$ 10 (1 k Ω)				
Resistance	R $\times$ 100 (10 k Ω)				
Resistance	R $\times$ 1K (100 k Ω)				
AC Voltage	2.5V — Variac direct				
AC Voltage	10V — Variac direct				
AC Voltage	50V — Variac direct				
AC Voltage	250V — UPS step-up				
AC Voltage	500V — UPS step-up				
AC Voltage	1000V — max ~975V			97.5% range	Noted
AC Voltage	5000V	Beyond Variac limit	—	—	Not verified

Notes / Observations:

■ @TheConfigBench — 4-Part Episode Series: Ep 1 — Theory: D'Arsonval meter movements, Series 8 internals, what the 4 pots do. Ep 2 — Build the Dim Bulb Tester: Parts, wiring, bypass switch, why every bench needs one. Ep 3 — AC Calibration Station: Isolation Xfmr → Dim Bulb → Variac → UPS step-up transformer. Pre-test procedure live on camera. Measuring turns ratio with XDM1241. Ep 4 — Live Calibration: Full procedure, range by range, both Simpson 260s side by side.